

Hyperbola

Question1

If θ is the angle subtended by a latus rectum at the centre of the hyperbola having eccentricity $\frac{2}{\sqrt{7}-\sqrt{3}}$, then $\sin \theta =$

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Options:

A.

$$\frac{1}{2} \tan \frac{\theta}{2}$$

B.

$$2 \cos \frac{\theta}{2}$$

C.

$$\frac{1}{\sin \frac{\theta}{2} + \cos \frac{\theta}{2}}$$

D.

$$1 - \cos \frac{\theta}{2}$$

Answer: A

Solution:

The latus rectum of a hyperbola is a line segment that is perpendicular to the main axis (transverse axis) and passes through a focus.

The angle θ is the angle at the center of the hyperbola made by one latus rectum.

For a hyperbola, the angle θ subtended at the center by the latus rectum always follows this rule:

$$\sin \theta = \frac{1}{2} \tan \left(\frac{\theta}{2} \right)$$

Question2

The tangent drawn at an extremity (in the first quadrant) of latus rectum of the hyperbola $\frac{x^2}{4} - \frac{y^2}{5} = 1$ meets the X -axis and Y -axis at A and B respectively. If O is the origin, then $(OA)^2 - (OB)^2 =$

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Options:

A.

$$-\frac{20}{9}$$

B.

$$\frac{16}{9}$$

C.

$$-\frac{4}{9}$$

D.

$$-\frac{4}{3}$$

Answer: A

Solution:

$$\frac{x^2}{4} - \frac{y^2}{5} = 1$$

$$\text{So, } a = 2, b = \sqrt{5} \text{ and } e = \sqrt{1 + \frac{5}{4}} = 3/2$$

$$\text{foci are } (\pm ae, 0) = (\pm 2 \times \frac{3}{2}, 0) = (\pm 3, 0)$$

The latus rectum for the focus in the positive x -direction is at $x = ac = 3$

$$\therefore \frac{3^2}{4} - \frac{y^2}{5} = 1 \Rightarrow y = \pm 5/2$$

in first quadrant, the extremity is $(3, 5/2)$ and the equation of tangential $(3, 5/2)$ is

$$\frac{x \cdot 3}{4} - \frac{y \cdot \frac{5}{2}}{5} = 1 \Rightarrow \frac{3x}{4} - \frac{y}{2} = 1$$

$$\Rightarrow 3x - 2y = 4$$

the tangent meet X -axis at $y = 0$

$$\Rightarrow 3x = 4 \Rightarrow x = 4/3$$

$$\therefore (OA)^2 = \left(\frac{4}{3} - 0\right)^2 + (0 - 0)^2 = 16/9$$

$$\text{and when } x = 0 \Rightarrow -2y = 4 \Rightarrow y = -3$$

$$\text{So, } OB^2 = (0 - 0)^2 + (0 + 2)^2 = 4$$

$$\text{Hence, } (OA)^2 - (OB)^2 = \frac{16}{9} - 4 = -20/9$$

Question3

Let $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$, where $\theta + \phi = \frac{\pi}{2}$ be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ If (h, k) is the point of intersection of the normals drawn at P and Q then $K =$

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Options:

A.

$$\frac{a^2+b^2}{a}$$

B.

$$-\left(\frac{a^2+b^2}{b}\right)$$

C.

$$-\left(\frac{a^2+b^2}{a}\right)$$

D.

$$\frac{a^2+b^2}{b}$$

Answer: B

Solution:

$P(a \sec \theta, b \tan \theta)$ and

$Q(a \sec \phi, b \tan \phi)$

$$\therefore \theta + \phi = \pi/2$$

Equation of normal at $(a \sec \theta, b \tan \theta)$ to hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$ax \cos \theta + by \cot \theta = a^2 + b^2 \quad \dots (i)$$

$$ax \cos \phi + by \cot \phi = a^2 + b^2$$

$$\Rightarrow \phi = \frac{\pi}{2} - \theta$$

$$\therefore ax \sin \theta + by \tan \theta = a^2 + b^2 \quad \dots (ii)$$

Multiply Eq. (i) by $\sin \theta$ and Eq. (ii) $\cos \theta$ and subtract

$$\text{by } \cot \theta \cdot \sin \theta - \text{by } \tan \theta \cos \theta$$

$$= (a^2 + b^2)(\sin \theta - \cos \theta)$$

$$\text{by } (\cos \theta - \sin \theta) = -(a^2 + b^2)$$

$$\Rightarrow y = -\frac{a^2 + b^2}{b}$$

Question4

If the angle between the asymptotes of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $2 \tan^{-1} \left(\frac{2}{3} \right)$ and $a^2 - b^2 = 45$, then $ab =$

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Options:

A.

20

B.

24

C.

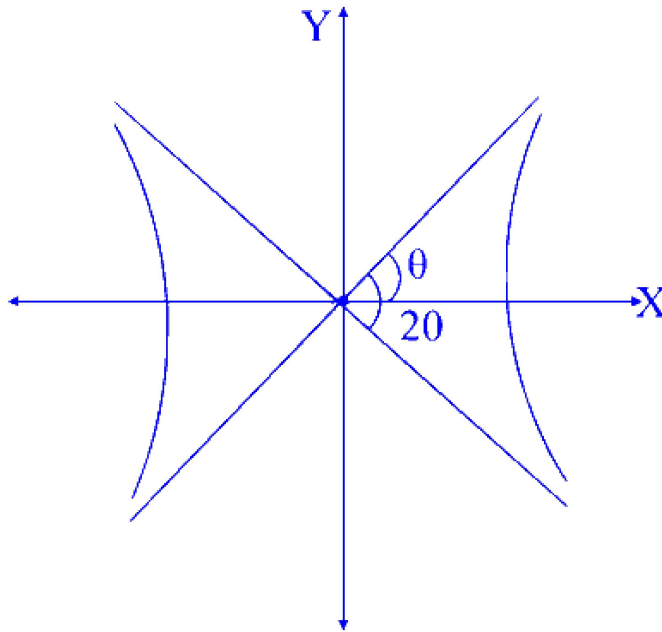
45

D.

54

Answer: D

Solution:



$$\therefore 2\theta = 2 \tan^{-1} 2/3$$

$$\Rightarrow \theta = \tan^{-1} 2/3$$

$$\Rightarrow \tan \theta = 2/3$$

$$\therefore e = \sec \theta$$

$$\Rightarrow e^2 = \sec^2 \theta = 1 + \frac{4}{9} = \frac{13}{9}$$

$$\therefore e^2 = \frac{13}{9}$$

$$\Rightarrow 1 + \frac{b^2}{a^2} = \frac{13}{9} \Rightarrow \frac{b^2}{a^2} = \frac{4}{9}$$

$$\Rightarrow \frac{b}{a} = \frac{2}{3} \quad \dots (i)$$

$$\Rightarrow a^2 - b^2 = 45 \Rightarrow a^2 - \frac{4a^2}{9} = 45$$

$$\Rightarrow 5a^2 = 9 \times 45$$

$$\Rightarrow a^2 = 81$$

$$\Rightarrow a = 9 \Rightarrow b = \frac{2}{3} \times 9 = 6$$

$$\therefore ab = 9 \times 6 = 54$$

Question5

If $3\sqrt{2}x - 4y = 12$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{5}{4}$ is its eccentricity, then $a^2 - b^2 =$

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Options:

A.

5

B.

7

C.

9

D.

11

Answer: B

Solution:

Given, equation of hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Now, the equation of the tangent to be hyperbola is

$$3\sqrt{2}x - 4y = 12$$

$$\Rightarrow y = \frac{3\sqrt{2}}{4}x - 3$$

So, slope of the tangent, $m_t = \frac{3\sqrt{2}}{4}$ and y -intercept is $c = -3$

Now, the line $y = mx + c$ tangent to the hyperbola is given by

$$c^2 = a^2m^2 - b^2$$

$$\Rightarrow (-3)^2 = a^2\left(\frac{3\sqrt{2}}{4}\right)^2 - b^2$$

$$\Rightarrow 9 = \frac{9}{8}a^2 - b^2$$

$$\Rightarrow 9a^2 - 8b^2 = 72 \quad \dots (i)$$

Given, eccentricity, $e = \frac{5}{4}$

$$\begin{aligned} \text{So, } b^2 &= a^2 (e^2 - 1) \\ &= a^2 \left(\left(\frac{5}{4} \right)^2 - 1 \right) \\ &= a^2 \left(\frac{25}{16} - 1 \right) = \frac{9}{16} a^2 \\ \Rightarrow 16b^2 &= 9a^2 \quad \dots (ii) \end{aligned}$$

From Eqs. (i) and (ii), we get

$$\begin{aligned} b^2 &= 9 \text{ and } a^2 = 16 \\ \therefore a^2 - b^2 &= 16 - 9 = 7 \end{aligned}$$

Question6

If the normal drawn to the hyperbola $xy = 16$ at $(8, 2)$ meets the hyperbola again at a point (α, β) , then $|\beta| + \frac{1}{|\alpha|} =$

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Options:

A.

40

B.

34

C.

28

D.

54

Answer: B

Solution:

Given hyperbola equation is $xy = 16$

Differentiate it w.r.t x , we get

$$y + x \cdot \frac{dy}{dx} = 0$$
$$\Rightarrow \frac{dy}{dx} = \frac{-y}{x}$$

Now, slope of tangent at $(8, 2)$ is

$$m_1 = \frac{-2}{8} = \frac{-1}{4}$$

And slope of normal is

$$m_n = \frac{-1}{-1/4} = 4$$

Now, equation of normal at $(8, 2)$ is

$$y - 2 = 4(x - 8)$$
$$\Rightarrow y = 4x - 30$$

Normal intersects the hyperbola again at a point (α, β) ,

$$\text{So, } \alpha\beta = 16 \text{ and } \beta = 4\alpha - 30$$

Solving these two equations, we get

$$\alpha(4\alpha - 30) = 16$$
$$\Rightarrow 2\alpha^2 - 15\alpha - 8 = 0$$

$$So, \alpha = \frac{-(-15) \pm \sqrt{(-15)^2 - 4 \cdot 2 \cdot (-8)}}{2 \cdot 2}$$
$$= \frac{15 \pm \sqrt{289}}{4} = \frac{15 \pm 17}{4}$$

$$\text{So, } \alpha_1 = \frac{15+17}{4} = \frac{32}{4} = 8 \text{ and}$$

$$\alpha_2 = \frac{15-17}{4} = \frac{-1}{2}$$

Since $(8, 2)$ is one point, the other point is where $\alpha = \frac{-1}{2}$

$$\text{So, } \beta = 4\alpha - 30$$
$$= 4 \left(\frac{-1}{2} \right) - 30 = -32$$

$$\therefore |\beta| + \frac{1}{|\alpha|} = |-32| + \frac{1}{|-1/2|}$$
$$= 32 + 2 = 34$$

Question 7

If $3x + 2\sqrt{2}y + k = 0$ is a normal to the hyperbola $4x^2 - 9y^2 - 36 = 0$ making positive intercepts on both the axes, then $k =$

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Options:

A.

$$13\sqrt{2}$$

B.

$$-5\sqrt{2}$$

C.

$$-2\sqrt{2}$$

D.

$$-13\sqrt{2}$$

Answer: D

Solution:

$$4x^2 - 9y^2 = 36 \Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\text{So, } a^2 = 9, b^2 = 4$$

Thus, equation of normal

$$\frac{9x}{x_1} + \frac{4y}{y_1} = (9 + 4) = 13 \quad \dots (i)$$

and the given equation is

$$\begin{aligned} 3x + 2\sqrt{2}y + k &= 0 \\ \Rightarrow 3x + 2\sqrt{2}y &= -k \quad \dots (ii) \end{aligned}$$

$\therefore$ Both equations are proportional

$$\text{Thus, } \frac{\frac{9}{x_1}}{3} = \frac{\frac{4}{y_1}}{2\sqrt{2}} = \frac{13}{-k}$$

$$\Rightarrow \frac{3}{x_1} = \frac{\sqrt{2}}{y_1} \Rightarrow x_1 = \frac{3}{\sqrt{2}}y_1$$

Put in equation of hyperbola

$$\frac{\left(\frac{3}{\sqrt{2}}y_1\right)^2}{9} - \frac{y^2}{4} = 1 \Rightarrow y_1^2 = 4 \Rightarrow y_1 = \pm 2$$

$$\text{at } y_1 = 2 \Rightarrow x_1 = 3\sqrt{2} \text{ and at } y_1 = -2$$

$$\Rightarrow x_1 = -3\sqrt{2}$$

$$\text{Also, } \frac{3}{x_1} = -\frac{13}{k} \Rightarrow -k = \frac{13}{3} \times 3\sqrt{2} = 13\sqrt{2}$$

$$\text{Hence, } k = -13\sqrt{2}$$

Question8

If a hyperbola has asymptotes $3x - 4y - 1 = 0$ and $4x - 3y - 6 = 0$, then the transverse and conjugate axes of that hyperbola are

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Options:

A.

$$x + y - 5 = 0, x - y - 1 = 0$$

B.

$$4x - 3y = 0, 3x + 4y = 0$$

C.

$$3x - 4y = 0, 4x + 3y = 0$$

D.

$$x + 2y - 1 = 0, 2x - y + 1 = 0$$

Answer: A

Solution:

Asymptotes are

$$\begin{array}{ll} 3x - 4y - 1 = 0 & \dots (i) \\ 4x - 3y - 6 = 0 & \dots (ii) \end{array}$$

By Eqs. (i) and (ii), we get

$$x = 3, y = 2$$

Now, the direction vectors of asymptotes for $3x - 4y = 1 \Rightarrow d_1 = \langle 4, 3 \rangle$ and for $4x - 3y = 6 \Rightarrow d_2 = \langle 3, 4 \rangle$ Thus, transverse and conjugate axis directions.

$$\begin{aligned} \frac{L_1}{\sqrt{a_1^2 + b_1^2}} &= \pm \frac{L_2}{\sqrt{a_2^2 + b_2^2}} \\ \Rightarrow \frac{3x - 4y - 1}{\sqrt{3^2 + (-4)^2}} &= \pm \frac{4x - 3y - 6}{\sqrt{4^2 + (-3)^2}} \\ \Rightarrow 3x - 4y - 1 &= \pm 4x - 3y - 6 \\ \text{So, } 3x - 4y - 1 &= 4x - 3y - 6 \\ \text{or } 3x - 4y - 1 &= -4x + 3y + 6 \\ \Rightarrow x + y - 5 &= 0 \\ \text{or } 7x - 7y - 7 &= 0 \\ \Rightarrow x + y - 5 &= 0 \text{ or } x - y - 1 = 0 \end{aligned}$$

Hence, transverse and conjugate axis are
 $x + y = 5, x - y = 1$

Question9

$x + y + 3 = 0, 2x - y + 1 = 0$ are the equations of the asymptotes of a hyperbola.

If $(1, -2)$ is a point on this hyperbola, then the equation of its conjugate hyperbola is

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Options:

A.

$$2x^2 + xy - y^2 + 7x - 2y - 1 = 0$$

B.

$$2x^2 + xy - y^2 + 7x - 2y + 13 = 0$$

C.

$$2x^2 + xy + y^2 - 7x - 2y - 1 = 0$$

D.

$$2x^2 + xy + y^2 - 7x - 2y + 13 = 0$$

Answer: B

Solution:

Given equation of asymptotes

$$x + y = -3 \quad \dots (i)$$

$$\text{and } 2x - y = -1 \quad \dots (ii)$$

On solving Eqs. (i) and (ii), we get centre of hyperbola i.e. $x = \frac{-4}{3}, y = \frac{-5}{3}$

$$\text{i.e. Centre} \equiv \left(\frac{-4}{3}, \frac{-5}{3}\right)$$

Also combined equation of asymptotes is $(x + y + 3)(2x - y + 1) = 0$

$$\Rightarrow 2x^2 - y^2 + xy + 7x - 2y + 3 = 0$$

Now, equation of hyperbola is (when equation of asymptotes are known) is

$$(x + y + 3)(2x - y + 1) + c = 0 \quad \dots (iii)$$

where c is constant.

$\therefore$ (iii) passes through $(1, -2)$.

$$\text{So, } (1 - 2 + 3)(2 + 2 + 1) + c = 0$$

$$\Rightarrow (2 \times 5) + c = 0 \Rightarrow c = -10$$

$\therefore$ Equation of hyperbola is

$$(x + y + 3)(2x - y + 1) - 10 = 0$$

Also, equation of conjugate hyperbola is (as we know that, equation of hyperbola, conjugate hyperbola, and pair of asymptotes are same but only difference in constant term).

$$= 2x^2 + xy - y^2 + 7x - 2y + 13 = 0$$

Question10

If θ is the acute angle between the tangents drawn from the point $(1, 1)$ to the hyperbola $4x^2 - 5y^2 - 20 = 0$, then $\tan \theta =$

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Options:

A.

$$2\sqrt{21}$$

B.

$$\frac{4}{5}$$

C.

$$\frac{\sqrt{7}}{2}$$

D.

$$\frac{2}{\sqrt{7}}$$

Answer: A

Solution:

Given equation of hyperbola can be written as

$$\frac{x^2}{5} - \frac{y^2}{4} = 1 \text{ which is of the form}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \dots (i)$$

$$\text{where, } a = \sqrt{5}, b = 2$$

As we know that, equation of any tangent to the hyperbola Eq. (i) having slope m is

$$y = mx \pm \sqrt{a^2m^2 - b^2} = mx + \sqrt{5m^2 - 4} \quad \dots (ii)$$

According to question, Eq. (ii) passes through $(1, 1)$.

$$\text{So, } 1 = m \times 1 \pm \sqrt{5m^2 - 4}$$

$$\Rightarrow 1 - m = \pm \sqrt{5m^2 - 4}$$

$$\Rightarrow (1 - m)^2 = 5m^2 - 4$$

$$\Rightarrow 4m^2 + 2m - 5 = 0 \quad \dots (iii)$$

Let m_1, m_2 slopes of these two tangents be the roots of Eq. (iii).

$$\text{So, } m_1 + m_2 = \frac{-2}{4} = \frac{-1}{2}$$

$$\begin{aligned}
 m_1 m_2 &= \frac{-5}{4} \\
 \therefore \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
 &= \left| \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \right| \\
 &= \left| \frac{\sqrt{\left(\frac{-1}{2}\right)^2 - 4 \times \left(\frac{-5}{4}\right)}}{1 + \left(\frac{-5}{4}\right)} \right| \\
 &= \left| \frac{\sqrt{\frac{1}{4} + \frac{20}{4}}}{1 - \frac{5}{4}} \right| = \left| \frac{\sqrt{21}}{2} \times \frac{4}{-1} \right| \\
 &= |-2\sqrt{21}| = 2\sqrt{21}
 \end{aligned}$$

Question11

If the equation of the tangent of the hyperbola $5x^2 - 9y^2 - 20x - 18y - 34 = 0$ which makes an angle 45° with the positive X -axis in positive direction is $x + by + c = 0$, then $b^2 + c^2 =$

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Options:

A.

2 or 13

B.

5 or 26

C.

2 or 26

D.

26 or 28

Answer: C

Solution:

The slope of tangent $m = \tan 45^\circ = 1$

$$\text{Now, } 5x^2 - 9y^2 - 20x - 18y - 34 = 10$$

$$\Rightarrow 5(x^2 - 4x) - 9(y^2 + 2y) = 34$$

$$\Rightarrow 5(x^2 - 4x + 4) - 9(y^2 + 2y + 1) = 34 + 20 - 9$$

$$\Rightarrow 5(x - 2)^2 - 9(y + 1)^2 = 45$$

$$\Rightarrow \frac{(x - 2)^2}{9} - \frac{(y + 1)^2}{5} = 1$$

The equation of the tangent with slope m is

$$y + 1 = m(x - 2) \pm \sqrt{9m^2 - 5}$$

Since, $m = 1$

$$\Rightarrow y + 1 = (x - 2) \pm \sqrt{9 - 5}$$

$$\Rightarrow y = x - 1 \text{ or } y = x - 5$$

$$\Rightarrow \text{The equation } x - y - 1 = 0$$

$$\text{or } x - y - 5 = 0$$

Comparing with $x + by + c = 0$

$$\Rightarrow b = -1 \text{ and } c = -1, -5$$

$$\Rightarrow b^2 + c^2 = (-1)^2 + (-1)^2 = 1 + 1 = 2$$

$$\text{and } b^2 + c^2 = (-1)^2 + (-5)^2 = 1 + 25 = 26$$

$$\Rightarrow b^2 + c^2 = 2 \text{ or } 26$$

Question12

If the distance between the foci of a hyperbola H is 26 and distance between its directrices is $\frac{50}{13}$, then the eccentricity of the conjugate hyperbola of the hyperbola H is

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Options:

A.

$$\frac{13}{12}$$

B.

$$\frac{25}{17}$$

C.

$$\frac{13}{7}$$

D.

$$\frac{25}{13}$$

Answer: A

Solution:

Distance between the foci = $2ae$

$$\Rightarrow 2ae = 26$$

distance between the directrices = $\frac{2a}{e}$

$$\Rightarrow \frac{2a}{e} = \frac{50}{13}$$

$$\Rightarrow \frac{2ae}{\frac{2a}{e}} = \frac{26}{\frac{50}{13}}$$

$$\Rightarrow e^2 = \frac{13 \cdot 13}{25}$$

$$\Rightarrow e = \frac{13}{5}$$

$$\Rightarrow \frac{1}{e_H^2} + \frac{1}{e_{CH}^2} = 1$$

$$\Rightarrow \frac{25}{169} + \frac{1}{e_{CH}^2} = 1$$

$$\Rightarrow e_{CH}^2 = \frac{169}{144} \Rightarrow e_{CH} = \frac{13}{12}$$

Question13

By rotating the axes about the origin in anti-clockwise direction with certain angle, if the equation $x^2 + 4xy + y^2 = 1$ is transformed to

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = l, \text{ then } \sqrt{\frac{a^2+b^2}{a^2}} =$$

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Options:

A.

$$2$$

B.

$$\frac{\sqrt{13}}{3}$$

C.

$$\frac{3}{2}$$

D.

$$\sqrt{10}$$

Answer: A

Solution:

Given equation

$$x^2 + 4xy + y^2 = 1 \quad \dots (i)$$

$$ax^2 + 2hxy + by^2 = 1$$

$$x = X \cos \theta - Y \sin \theta$$

$$y = X \sin \theta + Y \cos \theta$$

$$a = 1, h = 2, b = 1$$

$$h^2 - ab = 3 > 0 \Rightarrow \text{Given}$$

curve and hyperbola,

$$\tan 2\theta = \frac{2h}{a-b} = \frac{4}{0} = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{4}$$

$$x = \frac{X-Y}{\sqrt{2}}, y = \frac{X+Y}{\sqrt{2}}$$

From (i), we get

$$\frac{X^2 + Y^2 - 2XY}{2} + \frac{4(X^2 - Y^2)}{2} + \frac{X^2 + Y^2 + 2XY}{2} = 1$$

$$\frac{1}{2} [6X^2 - 2Y^2] = 1 \Rightarrow 3X^2 - Y^2 = 1$$

$$\frac{X^2}{1/3} - \frac{Y^2}{1} = 1 \Rightarrow a^2 = \frac{1}{3}, b^2 = 1$$

$$\sqrt{\frac{a^2 + b^2}{a^2}} = \sqrt{\frac{\frac{1}{3} + 1}{\frac{1}{3}}} = 2$$

Question14

If a tangent to the hyperbola $xy = -1$ is also a tangent to the parabola $y^2 = 8x$, then the equation of that tangent is

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Options:

A.

$$3y + x = 2$$

B.

$$y = 3x + 4$$

C.

$$y = x + 2$$

D.

$$y = 2x + 1$$

Answer: C

Solution:

Given equation of hyperbola

$$xy = -1$$

Let equation of tangent to $xy = -1$ be $y = mx + c$

$$\begin{aligned}\Rightarrow x(mx + c) + 1 &= 0 \\ mx^2 + cx + 1 &= 0 \text{ for tangent } Q = 0 \\ c^2 - 4m &= 0 \Rightarrow c^2 = 4m \quad \dots (i)\end{aligned}$$

Equation of tangent to $y^2 = 8x$

$$\begin{aligned}y &= mx + \frac{2}{m} \\ \frac{2}{m} = c &\Rightarrow \frac{4}{m^2} = 4m \\ m &= 1, c = 2 \\ y &= x + 2\end{aligned}$$

Question15

The distance between the tangents of the hyperbola $2x^2 - 3y^2 = 6$ which are perpendicular to the line $x - 2y + 5 = 0$ is

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Options:

A.

$$2\sqrt{2}$$

B.

$$4$$

C.

$$\sqrt{2}$$

D.

$$3\sqrt{2}$$

Answer: A

Solution:

Given equation of hyperbola

$$2x^2 - 3y^2 = 6$$

$$\frac{x^2}{3} - \frac{y^2}{2} = 1 \quad \dots (i)$$

Equation of tangent, $y =$ Slope of give line $m_1 = \frac{1}{2}$

Slope of perpendicular line $= -2$

$y = mx + c$ is tangent to hyperbola

$$c^2 = a^2m^2 - b^2 \Rightarrow c = \pm\sqrt{10}$$

Equation of tangents $2x + y + \sqrt{10} = 0,$

$$2x + y - \sqrt{10} \text{ distance } = \frac{2\sqrt{10}}{\sqrt{5}} = 2\sqrt{2}$$

Question16

The tangents drawn to the hyperbola $5x^2 - 9y^2 = 90$ through a variable point P make the angles α and β with its transverse axis. If α, β are the complementary angles then the locus of P is

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Options:

A.

$$x^2 + y^2 = 8$$

B.

$$x^2 - y^2 = 8$$

C.

$$x^2 - y^2 = 28$$

D.

$$x^2 + y^2 = 28$$

Answer: C

Solution:

Given, hyperbola $5x^2 - 9y^2 = 90$

$$\frac{x^2}{18} - \frac{y^2}{10} = 1$$

Let equation of tangent to the ellipse be

$$y = mx \pm \sqrt{18m^2 - 10}$$

$$\text{or } (y - mx)^2 = 18m^2 - 10$$

$$\therefore y^2 + m^2x^2 - 2mxy - 18m^2 + 10 = 0$$

$$m^2(x^2 - 18) - 2mxy + y^2 + 10 = 0$$

Given tangent makes α and β angles with transverse axis are complementary

$$\therefore \alpha + \beta = \frac{\pi}{2}$$

$$\tan(\alpha + \beta) = \text{not defined}$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \text{not defined}$$

$$\text{And } m_1 = \tan \alpha \quad m_2 = \tan \beta$$

$$m_1 + m_2 = \frac{2xy}{x^2 - 18} \text{ and } m_1 m_2 = \frac{y^2 + 10}{x^2 - 18}$$

$$\therefore 1 - m_1 m_2 = 0 \Rightarrow m_1 m_2 = +1$$

$$\frac{y^2 + 10}{x^2 - 18} = +1$$

$$\Rightarrow y^2 + 10 = x^2 - 18$$

$$\Rightarrow y^2 = x^2 - 28 \text{ or } x^2 - y^2 = 28$$

Question17

If θ is the acute angle between the asymptotes of a hyperbola $7x^2 - 9y^2 = 63$, then $\cos \theta =$

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Options:

A.

$$\frac{1}{4}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{1}{8}$$

D.

$$\frac{4}{3}$$

Answer: C

Solution:

We have hyperbola

$$7x^2 - 9y^2 = 63$$

$$\frac{x^2}{9} - \frac{y^2}{7} = 1$$

Now, its asymptotes is

$$\frac{x}{3} \pm \frac{y}{\sqrt{7}} = 0$$

$$r = \frac{x^2}{9} + 0 \cdot xy - \frac{y^2}{7} = 0$$

$$a = \frac{1}{9}, b = -\frac{1}{7}$$

Angle between pair of asymptotes

$$\tan \theta = \frac{2\sqrt{0 + \frac{1}{63}}}{\frac{1}{9} - \frac{1}{7}} = \left| \frac{2\frac{1}{\sqrt{63}}}{\frac{-2}{63}} \right|$$

$$\Rightarrow \tan \theta = \frac{2}{\sqrt{63}} \times \frac{63}{2} = \sqrt{63}$$

$$\therefore \cos \theta = \frac{1}{8}$$

Question18

One of the latus recta of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ subtends an angle $2 \tan^{-1} \left(\frac{3}{2} \right)$ at the centre of the hyperbola. If $b^2 = 36$ and e is the eccentricity of the given hyperbola, then $\sqrt{a^2 + e^2} =$

Options:

A.

4

B.

$\sqrt{14}$

C.

6

D.

$\sqrt{21}$

Answer: A

Solution:

Given, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $b^2 = 36$ Coordinate of one of $e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{36}{a^2}$ The latus rectum is $L\left(ae, +\frac{b^2}{a}\right)$ and $L'\left(ae, -\frac{b^2}{a}\right)$.

Now, slope of $OL = \frac{b^2}{a^2e}$ Slope of $OL' = -\frac{b^2}{a^2e}$ Now, angle between OL and OL'

$$2 \tan^{-1} \frac{b^2}{a^2e} = 2 \tan^{-1} \frac{3}{2}$$

$$\therefore \frac{b^2}{a^2 + e} = \frac{3}{2} \Rightarrow \frac{3e}{2} = \frac{b^2}{a^2}$$

$$\frac{3e}{2} = e^2 - 1 \quad \left[\because e^2 = 1 + \frac{b^2}{a^2} \right]$$

$$2e^2 - 3e - 2 = 0$$

$$(e - 2)(2e + 1) = 10 \Rightarrow e = 2, e \neq -\frac{1}{2}$$

$$a^2 = \frac{2b^2}{3e} = \frac{2 \times 36}{3 \times 2} = 12 \quad [\because b^2 = 36]$$

$$\therefore \sqrt{a^2 + e^2} = \sqrt{12 + 4} = 4$$

Question19

If the equation of the hyperbola having $(8, 3), (0, 3)$ as foci and $\frac{4}{3}$ as eccentricity is $\frac{(x-\alpha)^2}{p} - \frac{(y-\beta)^2}{q} = 1$, then $p + q =$

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Options:

A.

$$\beta^2$$

B.

$$\alpha + \beta$$

C.

$$\alpha^2$$

D.

$$\alpha\beta$$

Answer: C

Solution:

Given, foci $(8, 3)$ and $(0, 3)$ and $e = \frac{4}{3}$ centre $= (4, 3)$

Distance between foci $= 2ae = 8$

$$ae = 4$$

$$a \cdot \frac{4}{3} = 4$$

$$a = 3 \Rightarrow a^2 = 9 \text{ and } b^2 = 7$$

$\therefore$ Equation of hyperbola is

$$\frac{(x-4)^2}{9} - \frac{(y-3)^2}{7} = 1$$

Clearly $\alpha = 4, p = 9$ and $q = 7$

$$\therefore p + q = 16 = \alpha^2$$

Question20

If $y = x + \sqrt{2}$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2} = 1$, then equations of its directrices are

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Options:

A. $x = \pm\sqrt{3}$

B. $x = \pm\sqrt{\frac{8}{3}}$

C. $x = \pm\sqrt{\frac{2}{3}}$

D. $x = \pm\sqrt{\frac{4}{3}}$

Answer: B

Solution:

To find the directrices of the hyperbola given by the equation $\frac{x^2}{a^2} - \frac{y^2}{2} = 1$, where the tangent line is $y = x + \sqrt{2}$, we proceed with the following steps:

The equation for the tangent line of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ can be written as:

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

For our hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2} = 1$, $b^2 = 2$, and the given tangent line is $y = x + \sqrt{2}$, implying that the slope $m = 1$. Substituting these values, we find:

$$a^2m^2 - 2 = 2$$

$$a^2 \times 1^2 - 2 = 2 \implies a^2 - 2 = 2 \implies a^2 = 4$$

Next, we determine the eccentricity e of the hyperbola, given by:

$$e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{2}{4}} = \sqrt{\frac{3}{2}}$$

The equation of the directrices is given by:

$$x = \pm \frac{a}{e}$$

Substituting the known values:

$$x = \pm \frac{2}{\sqrt{3/2}} = \pm \frac{2 \cdot \sqrt{2}}{\sqrt{3}} = \pm \sqrt{\frac{8}{3}}$$

Thus, the directrices of the hyperbola are $x = \pm\sqrt{\frac{8}{3}}$.

Question21

The area of the quadrilateral formed with the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and its conjugate hyperbola is (in sq units)

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Options:

A. 24

B. 25

C. 16

D. 50

Answer: D

Solution:

We have the hyperbola given by $\frac{x^2}{16} - \frac{y^2}{9} = 1$. In this equation, $a^2 = 16$ and $b^2 = 9$.

To find the area of the quadrilateral formed with the foci of this hyperbola and its conjugate hyperbola, we use the formula:

$$\text{Area} = 2(a^2 + b^2)$$

Substituting the known values:

$$\text{Area} = 2(16 + 9) = 50 \text{ square units.}$$

Question22

The line $21x + 5y = k$ touches the hyperbola $7x^2 - 5y^2 = 232$, then k is equal to

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Options:

A. 116

B. 232

C. $58'$

D. 110

Answer: A

Solution:

To determine the value of k for which the line $21x + 5y = k$ is tangent to the hyperbola $7x^2 - 5y^2 = 232$, we apply the condition for tangency of a line to a hyperbola.

Tangency Condition:

For a line $y = mx + c$ to be tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the condition is $c^2 = a^2m^2 - b^2$.

Given:

The line equation is:

$$y = \frac{-21}{5}x + \frac{k}{5}$$

Hence, $m = \frac{-21}{5}$ and $c = \frac{k}{5}$.

The hyperbola $7x^2 - 5y^2 = 232$ can be written as:

$$\frac{x^2}{\frac{232}{7}} - \frac{y^2}{\frac{232}{5}} = 1$$

Here,

$$a^2 = \frac{232}{7}, \quad b^2 = \frac{232}{5}$$

Applying the Tangency Condition:

$$\left(\frac{k}{5}\right)^2 = \frac{232}{7} \times \left(\frac{-21}{5}\right)^2 - \frac{232}{5}$$

Simplifying:

$$\frac{k^2}{25} = \frac{232}{7} \times \frac{441}{25} - \frac{232}{5}$$

$$\frac{k^2}{25} = \frac{232 \times 441}{175} - \frac{232 \times 5}{25}$$

Factor out 232:

$$\frac{k^2}{25} = \frac{232 \times (441 - 5)}{175}$$

Calculate:

$$= \frac{232 \times 436}{175}$$

Now solving for k :

$$k^2 = \frac{13456}{25}$$

Therefore:

$$k = 116$$

Thus, the value of k is 116.

Question23

If the equation $\frac{x^2}{7-k} + \frac{y^2}{5-k} = 1$ represents a hyperbola, then

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Options:

A.

$$5 < k < 7$$

B. $k < 5$ or $k > 7$

C. $k < 5$

D.

$$k \neq 5, \quad k \neq 7, \quad -\infty < k < \infty$$

Answer: A

Solution:

Given, equation $\frac{x^2}{7-k} + \frac{y^2}{5-k} = 1$ represent at hyperbola.

We know that equation of hyperbola,

$$\frac{x^2}{a} - \frac{y^2}{b} = 1$$

$$\therefore a^2 > 0 \text{ and } b^2 > 0$$

$$\Rightarrow 7 - k > 0 \text{ and } k - 5 > 0$$

$$\Rightarrow k < 7 \text{ and } k > 5$$

$$\Rightarrow 5 < k < 7$$

Question24

The transformed equation of $x^2 - y^2 + 2x + 4y = 0$ when the origin is shifted to the point $(-1, 2)$ is

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Options:

A. $x^2 + y^2 = 1$

B. $x^2 + 3y^2 = 1$

C. $x^2 - y^2 + 3 = 0$

D. $4x^2 + 9y^2 = 36$

Answer: C

Solution:

To transform the equation $x^2 - y^2 + 2x + 4y = 0$ when the origin is shifted to the point $(-1, 2)$, we proceed as follows:

Given equation:

$$x^2 - y^2 + 2x + 4y = 0$$

New origin:

$$(h, k) = (-1, 2)$$

Transform the variables:

$$\text{Let } X = x - h \text{ and } Y = y - k.$$

So,

$$X = x - (-1) = x + 1$$

$$Y = y - 2$$

Solving for x and y :

$$x = X - 1$$

$$y = Y + 2$$

Now, substituting these into the given equation:

$$(x - 1)^2 - (y + 2)^2 + 2(x - 1) + 4(y + 2) = 0$$

Expanding each term:

$$(x - 1)^2 = x^2 - 2x + 1$$

$$(y + 2)^2 = y^2 + 4y + 4$$

Substitute and simplify:

$$\begin{aligned}[x^2 - 2x + 1] - [y^2 + 4y + 4] + 2(x - 1) + 4(y + 2) &= 0 \\ x^2 - y^2 - 2x + 1 - 4y - 4 + 2x - 2 + 4y + 8 &= 0\end{aligned}$$

Combine and simplify:

$$x^2 - y^2 + 3 = 0$$

Thus, the transformed equation is:

$$x^2 - y^2 + 3 = 0$$

Question25

If the ellipse $4x^2 + 9y^2 = 36$ is confocal with a hyperbola whose length of the transverse axis is 2 , then the points of intersection of the ellipse and hyperbola lie on the circle

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Options:

A. $x^2 + y^2 = 81$

B. $x^2 + y^2 = 16$

C. $x^2 + y^2 = 25$

D. $x^2 + y^2 = 5$

Answer: D

Solution:

Given the ellipse equation:

$$4x^2 + 9y^2 = 36$$

or equivalently,

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \dots (i)$$

This ellipse is confocal with a hyperbola whose transverse axis has a length of 2. For the hyperbola, the transverse axis is $2a = 2$, giving us $a = 1$.

For confocal conics, the foci are at the same distance from the origin. The distance of the focus for the ellipse is:

$$c = \sqrt{a^2 - b^2} = \sqrt{\sqrt{9^2} - \sqrt{4^2}} = \sqrt{9 - 4} = \sqrt{5}$$

For the hyperbola, we have:

$$c = \sqrt{a^2 + b^2}$$

Thus,

$$c = \sqrt{1 + b^2} = \sqrt{5}$$

Solving for b^2 , we find $b^2 = 4$. Therefore, the equation of the hyperbola is:

$$\frac{x^2}{1} - \frac{y^2}{4} = 1 \quad \Rightarrow \quad x^2 - \frac{y^2}{4} = 1 \quad \dots \text{(ii)}$$

This implies:

$$y^2 = 4(x^2 - 1) \quad \dots \text{(iii)}$$

From equations (i) and (iii), we get:

$$4x^2 + 9(4(x^2 - 1)) = 36$$

$$4x^2 + 36x^2 - 36 = 36$$

$$40x^2 = 72$$

$$x^2 = \frac{9}{5}$$

$$x = \pm \frac{3\sqrt{5}}{5}$$

Thus, using equation (iii):

$$y^2 = 4 \times \left(\frac{9}{5} - 1 \right) = 4 \times \frac{4}{5} = \frac{16}{5}$$

$$y = \pm \frac{4\sqrt{5}}{5}$$

The points of intersection are $\left(\pm \frac{3\sqrt{5}}{5}, \frac{4\sqrt{5}}{5} \right)$.

The equation of a circle passing through these points is:

$$\begin{aligned} x^2 + y^2 &= \left(\frac{3\sqrt{5}}{5} \right)^2 + \left(\frac{4\sqrt{5}}{5} \right)^2 \\ &= \frac{45 + 80}{25} = \frac{125}{25} \end{aligned}$$

$$\therefore x^2 + y^2 = 5$$

Question 26

If the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\sec \alpha$, then area of the triangle formed by the asymptotes of the hyperbola with any of its tangent is

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Options:

A. $a^2 b^2 \sec^2 \alpha$

B. $\frac{b^2}{|\tan \alpha|}$

C. $a^2 \tan^2 \alpha$

D. $(a^2 + b^2) \tan^2 \alpha$

Answer: B

Solution:

For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the eccentricity e is given by the formula:

$$e = \sqrt{1 + \frac{b^2}{a^2}}$$

Since the eccentricity is provided as $\sec \alpha$, we have:

$$\sec \alpha = \sqrt{1 + \frac{b^2}{a^2}}$$

Squaring both sides of the equation, we obtain:

$$\sec^2 \alpha = 1 + \frac{b^2}{a^2}$$

Subtracting 1 from both sides gives:

$$\sec^2 \alpha - 1 = \frac{b^2}{a^2} = \tan^2 \alpha$$

or

$$\frac{b^2}{a^2} = \tan^2 \alpha$$

The equations of the asymptotes of the hyperbola are:

$$y = \pm \frac{b}{a} x = \pm \tan \alpha \cdot x$$

A general form for the tangent to the hyperbola is:

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Given $x_1 = a \sec \theta$ and $y_1 = b \tan \theta$, when $x = 0$, the equation transforms to:

$$\frac{-y \tan \theta}{a \tan \alpha} = 1 \Rightarrow y = -\frac{a \tan \alpha}{\tan \theta}$$

Thus, the vertices of the triangle formed by asymptotes and the tangent will be at $(\pm a, 0)$ and $(0, -\frac{a \tan \alpha}{\tan \theta})$.

The area of the triangle, with the base as $2a$ and height $-\frac{a \tan \alpha}{\tan \theta}$, is calculated by:

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{Height} = \frac{1}{2} \times 2a \times \left(\frac{a \tan \alpha}{\tan \theta} \right)$$

Simplifying this, we get:

$$= \left| \frac{a^2 \tan \alpha}{\tan \theta} \right| = \left| \frac{a^2 b^2}{a^2 \tan \theta} \right|$$

From our earlier identity, this equals:

$$\text{Area} = \frac{b^2}{\tan \theta} = \frac{b^2}{|\tan \alpha|}$$

Question 27

If e_1 and e_2 are respectively the eccentricities of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and its conjugate hyperbola, then the line $\frac{x}{2e_1} + \frac{y}{2e_2} = 1$ touches the circle having centre at the origin, then its radius is

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Options:

- A. 2
- B. $e_1 + e_2$
- C. $e_1 e_2$
- D. 4

Answer: A

Solution:

Given the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, its eccentricity e_1 is defined by:

$$e_1 = \sqrt{1 + \frac{b^2}{a^2}}$$

For the conjugate hyperbola $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$, the eccentricity e_2 is:

$$e_2 = \sqrt{1 + \frac{a^2}{b^2}}$$

We are given the line:

$$\frac{x}{2e_1} + \frac{y}{2e_2} = 1$$

Rewriting it as:

$$\frac{x}{2e_1} + \frac{y}{2e_2} - 1 = 0$$

Our task is to find the radius of the circle, centered at the origin, which this line touches.

The distance from the origin $(0, 0)$ to a line $Ax + By + C = 0$ is:

$$d = \frac{|C|}{\sqrt{A^2 + B^2}}$$

For equation (ii), the distance becomes:

$$d = \frac{|-1|}{\sqrt{\left(\frac{1}{2e_1}\right)^2 + \left(\frac{1}{2e_2}\right)^2}}$$

Substituting, we find:

$$d = \frac{2}{\sqrt{\frac{a^2}{a^2+b^2} + \frac{b^2}{a^2+b^2}}}$$

$$d = \frac{2}{\sqrt{\frac{a^2+b^2}{a^2+b^2}}} = \frac{2}{\sqrt{1}}$$

So, the distance d is 2. Therefore, the radius of the circle is 2.

Question28

The descending order of magnitude of the eccentricities of the following hyperbolas is

A. A hyperbola whose distance between foci is three times the distance between its directrices.

B. Hyperbola in which the transverse axis is twice the conjugate axis.

C. Hyperbola with asymptotes $x + y + 1 = 0$, $x - y + 3 = 0$

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Options:

A. C, A, B

B. B, C, A

C. C, B, A

D. A, C, B

Answer: D

Solution:

Explanation

Hyperbola A

$$\text{Given: } 2c = 3 \times \frac{2a}{e}$$

$$\text{This implies: } c = \frac{3a}{e}$$

Using the relation $c^2 = a^2 + b^2$, we derive:

$$\left(\frac{3a}{e}\right)^2 = a^2 + b^2$$

$$\Rightarrow \frac{9a^2}{e^2} = a^2 + b^2 \Rightarrow 9a^2 = e^2(a^2 + b^2)$$

$$\Rightarrow 9 = e^2\left(1 + \frac{b^2}{a^2}\right) \Rightarrow 9 = e^2(1 + e^2 - 1)$$

$$\Rightarrow e^4 = 9 \Rightarrow e = \sqrt{3}$$

Hyperbola B

Let $2a$ be the transverse axis and $2b$ be the conjugate axis, given:

$$2a = 2 \times 2b \Rightarrow a = 2b$$

Using the relation $c^2 = a^2 + b^2$, we derive:

$$c^2 = (2b)^2 + b^2 \Rightarrow c^2 = 5b^2 \Rightarrow c = \sqrt{5}b$$

Therefore:

$$e = \frac{\sqrt{5}b}{2b} = \frac{\sqrt{5}}{2} \quad \left[\because e = \frac{c}{a} \right]$$

Hyperbola C

Given the asymptotes equations:

$$x + y + 1 = 0 \Rightarrow y = -x - 1$$

$$\text{and } x - y + 3 = 0 \Rightarrow y = x + 3$$

The standard form of asymptotes for a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by:

$$y = \pm \frac{b}{a}x$$

The slopes of asymptotes are -1 and 1 , meaning:

$$\frac{b}{a} = \pm 1 \Rightarrow b = a$$

Using the relation $c^2 = a^2 + b^2$, we derive:

$$c^2 = a^2 + a^2 \Rightarrow c = \sqrt{2}a$$

Therefore:

$$e = \frac{\sqrt{2}a}{a} \Rightarrow e = \sqrt{2}$$

Conclusion

In descending order of the magnitudes of the eccentricities, we have $\sqrt{3} > \sqrt{2} > \frac{\sqrt{5}}{2}$. Therefore, the order is A, C, B .

Question29

The equation of the pair of asymptotes of the hyperbola $4x^2 - 9y^2 - 24x - 36y - 36 = 0$ is

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Options:

A. $2x^2 - xy - 3y^2 - 14x - 9y - 12 = 0$

B. $2x^2 - xy - 3y^2 - 2x + 3y = 0$

C. $2x^2 - 5xy + 3y^2 - 22x + 27y + 60 = 0$

D. $4x^2 - 9y^2 - 24x - 36y = 0$

Answer: D

Solution:

To determine the equation of the asymptotes for the hyperbola given by:

$$4x^2 - 9y^2 - 24x - 36y - 36 = 0$$

we first rearrange and complete the square:

Reorganize the terms:

$$(4x^2 - 24x) - (9y^2 + 36y) - 36 = 0$$

Complete the square for each set of terms:

$$4(x^2 - 6x) - 9(y^2 + 4y) - 36 = 0$$

Completing the square:

$$4(x^2 - 6x + 9 - 9) - 9(y^2 + 4y + 4 - 4) - 36 = 0$$

Simplifying:

$$4(x - 3)^2 - 9(y + 2)^2 - 36 = 0$$

Set the equation in standard hyperbola form:

$$4(x - 3)^2 - 9(y + 2)^2 = 36$$

$$\frac{(x-3)^2}{9} - \frac{(y+2)^2}{4} = 1$$

The asymptotes of a hyperbola in the standard form $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ are given by:

$$(y - k) = \pm \frac{b}{a}(x - h)$$

For our hyperbola:

$$h = 3, k = -2$$

$$a = 3, b = 2$$

Thus, the equations of the asymptotes are:

$$(y + 2) = \pm \frac{2}{3}(x - 3)$$

Expanding this gives:

First Asymptote:

$$y + 2 = \frac{2}{3}(x - 3)$$

$$3(y + 2) = 2(x - 3)$$

Simplifying:

$$3y + 6 = 2x - 6 \implies 2x - 3y = 12$$

Second Asymptote:

$$y + 2 = -\frac{2}{3}(x - 3)$$

$$3(y + 2) = -2(x - 3)$$

Simplifying:

$$3y + 6 = -2x + 6 \implies 2x + 3y = 0$$

After multiplying and expanding these, we recover the simplified asymptote equation:

$$4x^2 - 9y^2 - 24x - 36y = 0$$

Question30

The equation of one of the tangents drawn from the point $(0, 1)$ to the hyperbola $45x^2 - 4y^2 = 5$ is

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Options:

A. $4y + 5 = 0$

B. $3x + 4y - 4 = 0$

C. $5x - 6y + 6 = 0$

D. $9x - 2y + 2 = 0$

Answer: D

Solution:

The equation of one of the tangent drawn from the point $(0, 1)$ to the hyperbola $45x^2 - 4y^2 = 5$.

We can use the fact that the equation of the tangent to a hyperbola.

$Ax^2 + By^2 = C$ at point (x_1, y_1) is given

$$Ax_1x + By_1y = C$$

Here, $A = 45, B = -4, C = 5$

Let (x_1, y_1) be the point of tangency, the equation of the tangent line is

$$45x_1x - 4y_1y = 5$$

The line must, also pass through the point $(0, 1)$ substituting $(x = 0)$ and $(y = 1)$ into the tangent equation gives

$$45x_1(0) - 4y_1(1) = 5$$

$$\Rightarrow y_1 = \frac{-5}{4}$$

$$45x_1x - 4\left(\frac{-5}{4}\right)y = 5$$

$$9x_1x + y = 1$$

Using the hyperbola equation

$$\begin{aligned}
45x_1^2 - 4\left(\frac{25}{16}\right) &= 5 \\
\Rightarrow 45x_1^2 - \frac{25}{4} &= 5 \\
\Rightarrow 45x_1^2 &= 5 + \frac{25}{4} = \frac{20 + 25}{4} = \frac{45}{4} \\
\Rightarrow 45x_1^2 &= \frac{45}{4} \\
\Rightarrow x_1^2 &= \frac{1}{4} \Rightarrow x_1 = \pm \frac{1}{2} \\
\Rightarrow x_1 &= \frac{1}{2} \\
\Rightarrow 9\left(\frac{1}{2}\right)x + y &= 1 \\
\Rightarrow 9x + 2y &= 2 \\
\Rightarrow x_1 = \frac{-1}{2} \text{ and } 9\left(\frac{-1}{2}\right)x + y &= 1 \\
\Rightarrow -9x + 2y &= 2 \\
\Rightarrow 9x - 2y + 2 &= 0
\end{aligned}$$

Question31

If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is $\sqrt{5}x = 4$ and e is its eccentricity, then $e^2 =$

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Options:

A. $\frac{\sqrt{7}}{2}$

B. $\frac{7}{2}$

C. $\frac{35}{4}$

D. $2\sqrt{3}$

Answer: B

Solution:

To solve for e^2 where the hyperbola is given by $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the hyperbola passes through the point $(4, -2\sqrt{3})$, we can proceed as follows:

Plug the point into the hyperbola equation:

$$\frac{16}{a^2} - \frac{12}{b^2} = 1 \quad \dots (i)$$

Relate b^2 and a^2 with eccentricity e :

$$b^2 = a^2(e^2 - 1)$$

Given the directrix $\sqrt{5}x = 4$:

$$x = \frac{4}{\sqrt{5}} = \frac{a}{e}$$

Therefore,

$$a^2 = \frac{16}{5}e^2 \quad \dots (ii)$$

Substitute the expression for a^2 from (ii) into (i):

$$\frac{16}{\frac{16}{5}e^2} - \frac{12}{b^2} = 1$$

Substitute $b^2 = a^2(e^2 - 1)$:

$$\frac{5}{e^2} - \frac{12}{a^2(e^2 - 1)} = 1$$

Replace a^2 with $\frac{16}{5}e^2$:

$$\frac{5}{e^2} - \frac{12}{\frac{16}{5}e^2(e^2 - 1)} = 1$$

Simplify the equation:

$$\frac{5}{e^2} - \frac{60}{16e^2(e^2 - 1)} = 1$$

On further simplification:

$$\frac{5}{e^2} \left[1 - \frac{12}{16(e^2 - 1)} \right] = 1$$

Solve for e^2 :

Upon solving, we find that:

$$e^2 = \frac{7}{2}$$

This process allows us to determine that the value of e^2 is $\frac{7}{2}$.

Question32

If l_1 and l_2 are the lengths of the perpendiculars drawn from a point on the hyperbola $5x^2 - 4y^2 - 20 = 0$ to its asymptotes, then $\frac{l_1^2 l_2^2}{100} =$

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Options:

A. $\frac{20}{9}$

B. $\frac{16}{81}$

C. $\frac{4}{81}$

D. $\frac{2}{9}$

Answer: C

Solution:

Given hyperbola $5x^2 - 4y^2 - 20 = 0$

$$\Rightarrow 5x^2 - 4y^2 = 20$$

$$\Rightarrow \frac{5x^2}{20} - \frac{4y^2}{20} = \frac{20}{20}$$

$$\Rightarrow \frac{x^2}{4} - \frac{y^2}{5} = 1$$

Asymptotes are $x \pm \frac{2}{\sqrt{5}}y = 0$

Let $P(x_0, y_0)$ be any point on $S = 0$, Length of the perpendicular from P to $x + \frac{2}{\sqrt{5}}y = 0$ is

$$l_1 = \frac{\left| x_0 + \frac{2}{\sqrt{5}}y_0 \right|}{\sqrt{1 + \frac{4}{5}}} = \frac{\left| x_0 + \frac{2}{\sqrt{5}}y_0 \right|}{\sqrt{\frac{9}{5}}} = \frac{\left| \sqrt{5}x_0 + 2y_0 \right|}{3}$$

Length of the perpendicular from P to $x - \frac{2}{\sqrt{5}}y = 0$ is

$$l_2 = \frac{\left| x_0 - \frac{2}{\sqrt{5}}y_0 \right|}{\sqrt{1 + \frac{4}{5}}} = \frac{\left| x_0 - \frac{2}{\sqrt{5}}y_0 \right|}{\sqrt{\frac{9}{5}}} = \frac{\left| \sqrt{5}x_0 - 2y_0 \right|}{3}$$

$$\therefore l_1^2 \cdot l_2^2 = \frac{\left[\left(\sqrt{5}x_0 + 2y_0 \right)^2 \left(\sqrt{5}x_0 - 2y_0 \right)^2 \right]}{(3)^2 \times (3)^2}$$

$$= \frac{(5x_0 - 4y_0)^2}{81} = \frac{400}{81}$$

$$\therefore \frac{l_1^2 l_2^2}{100} = \frac{400}{81 \times 100} = \frac{4}{81}$$

Question33

If a circle of radius 4 cm passes through the foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{4} = 1$ and concentric with the hyperbola, then the eccentricity of the conjugate hyperbola of that hyperbola is

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Options:

A. 2

B. $2\sqrt{3}$

C. $2/\sqrt{3}$

D. $\sqrt{3}$

Answer: A

Solution:

The eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{4} = 1$ is given by the formula:

$$\sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{4}{a^2}}$$

The center of this hyperbola is at the origin, (0, 0).

The equation of the circle is:

$$x^2 + y^2 = 16$$

This circle passes through the foci of the hyperbola, which are located at $(ae, 0)$. Thus, we can write:

$$a^2e^2 = 16$$

Since $e = \sqrt{1 + \frac{4}{a^2}}$, we substitute:

$$a^2 \left(1 + \frac{4}{a^2}\right) = 16$$

Expanding and simplifying gives:

$$a^2 + 4 = 16 \Rightarrow a^2 = 12$$

Further simplifying the eccentricity:

$$e = \sqrt{1 + \frac{4}{12}} = \frac{2}{\sqrt{3}}$$

Now, let e' be the eccentricity of the conjugate hyperbola. We have:

$$\frac{1}{e^2} + \frac{3}{4} = 1$$

Solving for e' :

$$\frac{1}{e'^2} = 1 - \frac{3}{4} \Rightarrow e'^2 = 4 \Rightarrow e' = 2$$

Question34

If a tangent to the hyperbola $x^2 - \frac{y^2}{3} = 1$ is also a tangent to the parabola $y^2 = 8x$, then equation of such tangent with the positive slope is

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Options:

- A. $y - x - \frac{1}{2} = 0$
- B. $y - 2x - 1 = 0$
- C. $2y - 4x - 1 = 0$
- D. $y - x - 1 = 0$

Answer: B

Solution:

To determine the equation of the tangent line that is tangent to both the hyperbola $x^2 - \frac{y^2}{3} = 1$ and the parabola $y^2 = 8x$, we start with the tangent equation for the parabola. The general equation for a tangent to the parabola $y^2 = 8x$ is:

$$y = mx + \frac{2}{m}$$

Since this line is also tangent to the hyperbola, it must satisfy the condition for tangency to the hyperbola as well. The condition for tangency to a hyperbola of the form $x^2 - \frac{y^2}{a^2} = 1$ is given by:

$$c = \pm\sqrt{m^2 - 3}$$

Given $c = \frac{2}{m}$, equating this to the tangency condition gives:

$$\frac{2}{m} = \pm\sqrt{m^2 - 3}$$

Squaring both sides, we get:

$$\frac{4}{m^2} = m^2 - 3$$

Rearranging terms gives:

$$m^4 - 3m^2 - 4 = 0$$

We can factor this as:

$$(m^2 - 4)(m^2 + 1) = 0$$

Solving for m^2 , we find:

$$m^2 = 4 \quad (\text{since } m^2 \neq -1)$$

Thus, $m = \pm 2$. For the tangent with a positive slope:

$$m = 2$$

Substituting $m = 2$ back into the tangent equation of the parabola, we have:

$$y = 2x + \frac{2}{2} = 2x + 1$$

So, the equation of the tangent line with a positive slope is:

$$y - 2x - 1 = 0$$

Question35

The locus of the mid-points of the chords of the hyperbola $x^2 - y^2 = a^2$ which touch the parabola $y^2 = 4ax$ is

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Options:

A. $x(y^2 - x^2) = ay^2$

B. $x(x^2 + y^2) = y^2 + x$

C. $ax^3 + y^3 = 3x$

D. $x(x^2 - y^2) = a^2$

Answer: A

Solution:

To find the locus of the midpoints of the chords of the hyperbola $x^2 - y^2 = a^2$ that touch the parabola $y^2 = 4ax$, we begin by considering a point (h, k) , which is the midpoint of a chord of the hyperbola.

The equation of this chord in the format $T = S_1$ is:

$$hx - ky = h^2 - k^2$$

Rewriting it in slope-intercept form gives:

$$y = \frac{h}{k}x + \frac{k^2 - h^2}{k}$$

Since this line is tangent to the parabola $y^2 = 4ax$, we use the point of tangency conditions:

For a line $y = mx + c$ tangent to the parabola, the condition is:

$$c = \frac{a}{m}$$

Substituting $m = \frac{h}{k}$ and $c = \frac{k^2 - h^2}{k}$, we equate and simplify:

$$\frac{k^2 - h^2}{k} = \frac{ak}{h}$$

Solving this equation gives:

$$hk^2 - h^3 = ak^2$$

Thus, the required locus of the midpoint (h, k) satisfies:

$$x(y^2 - x^2) = ay^2$$

Question36

If the product of eccentricities of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = -1$ is 1, then $b^2 =$

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Options:

A. $\frac{12}{25}$

B. 144

C. 25

D. $\frac{144}{25}$

Answer: D

Solution:

To find the value of b^2 , we need to consider the properties of the given ellipse and hyperbola. Let's denote e_1 and e_2 as the eccentricities of the ellipse and hyperbola, respectively.

For the ellipse given by $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$, the eccentricity e_1 is calculated as:

$$e_1 = \sqrt{1 - \frac{b^2}{16}}$$

For the hyperbola given by $\frac{x^2}{9} - \frac{y^2}{16} = -1$, the eccentricity e_2 is calculated as:

$$e_2 = \sqrt{1 + \frac{9}{16}}$$

We are given that the product of these eccentricities is 1:

$$e_1 \times e_2 = 1$$

Substituting the expressions for e_1 and e_2 :

$$\sqrt{1 - \frac{b^2}{16}} \times \sqrt{1 + \frac{9}{16}} = 1$$

Squaring both sides to eliminate the square roots, we get:

$$(1 - \frac{b^2}{16})(1 + \frac{9}{16}) = 1$$

Simplifying the equation:

$$\frac{(16-b^2)(9+16)}{16 \times 16} = 1$$

This further simplifies to:

$$16 - b^2 = \frac{256}{25}$$

Solving for b^2 :

$$b^2 = 16 - \frac{256}{25}$$

Putting the expression under a common denominator and simplifying gives us:

$$b^2 = \frac{400}{25} - \frac{256}{25} = \frac{144}{25}$$

Therefore, $b^2 = \frac{144}{25}$.

Question37

If the line $5x - 2y - 6 = 0$ is a tangent to the hyperbola $5x^2 - ky^2 = 12$, then the equation of the normal to this hyperbola at the point $(\sqrt{6}, p)$ ($p < 0$) is

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Options:

A. $\sqrt{6}x + 2y = 0$

B. $2\sqrt{6}x + 3y = 3$

C. $\sqrt{6}x - 5y = 21$

D. $3\sqrt{6}x - y = 21$

Answer: C

Solution:

Step 1: Find the value of k

The condition for the line $lx + my + n = 0$ to be a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } \frac{a^2 l^2}{n^2} - \frac{b^2 m^2}{n^2} = 1.$$

The given line is $5x - 2y - 6 = 0$, so $l = 5$, $m = -2$, $n = -6$.

The given hyperbola is $5x^2 - ky^2 = 12$, which can be written as $\frac{x^2}{12/5} - \frac{y^2}{12/k} = 1$. So

$$a^2 = \frac{12}{5} \text{ and } b^2 = \frac{12}{k}.$$

Substitute these into the tangency condition:

$$\frac{(12/5)(5^2)}{(-6)^2} - \frac{(12/k)(-2^2)}{(-6)^2} = 1$$

$$\frac{12 \times 25}{5 \times 36} - \frac{12 \times 4}{k \times 36} = 1$$

$$\frac{60}{36} - \frac{48}{36k} = 1$$

$$\frac{5}{3} - \frac{4}{3k} = 1$$

$$\frac{4}{3k} = \frac{5}{3} - 1 = \frac{2}{3}$$

$$k = \frac{4 \times 3}{2 \times 3} = 2$$

Step 2: Find the point of tangency

The hyperbola equation is $5x^2 - 2y^2 = 12$. The point $(\sqrt{6}, p)$ lies on the hyperbola, so:

$$5(\sqrt{6})^2 - 2p^2 = 12$$

$$5(6) - 2p^2 = 12$$

$$30 - 2p^2 = 12$$

$$2p^2 = 18$$

$$p^2 = 9$$

$$p = \pm 3$$

Given $p < 0$, we take $p = -3$. The point is $(\sqrt{6}, -3)$.

Step 3: Find the equation of the normal

The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at point (x_1, y_1) is

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2.$$

Using $a^2 = \frac{12}{5}$, $b^2 = \frac{12}{k} = \frac{12}{2} = 6$, $x_1 = \sqrt{6}$, $y_1 = -3$:

$$\frac{(12/5)x}{\sqrt{6}} + \frac{6y}{-3} = \frac{12}{5} + 6$$

$$\frac{12x}{5\sqrt{6}} - 2y = \frac{12 + 30}{5}$$

$$\frac{12x}{5\sqrt{6}} - 2y = \frac{42}{5}$$

Multiply by $5\sqrt{6}$ to clear denominators:

$$12x - 10\sqrt{6} y = 42\sqrt{6}$$

Divide by 2:

$$6x - 5\sqrt{6} y = 21\sqrt{6}$$

Rearrange to match the options:

$$6x - 5\sqrt{6} y - 21\sqrt{6} = 0$$

This is not exactly in the options format, let's recheck the options. The options are in the form of linear equations with coefficients. Option C is $\sqrt{6} x - 5y = 21$.

Let's re-examine the equation $\frac{12x}{5\sqrt{6}} - 2y = \frac{42}{5}$.

Divide by 2:

$$\frac{6x}{5\sqrt{6}} - y = \frac{21}{5}$$

Multiply by $5\sqrt{6}$:

$$6x - 5\sqrt{6} y = 21\sqrt{6}$$

Divide by $\sqrt{6}$:

$$\frac{6x}{\sqrt{6}} - 5y = 21$$

$$\sqrt{6} x - 5y = 21$$

Answer:

The correct option is (C) $\sqrt{6} x - 5y = 21$.

Question38

If the angle between the asymptotes of the hyperbola $x^2 - ky^2 = 3$ is $\frac{\pi}{3}$ and e is its eccentricity, then the pole of the line $x + y - 1 = 0$ with respect to this hyperbola is

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Options:

A. $\left(k, \frac{\sqrt{30}}{2}\right)$

B. $\left(-k, \frac{\sqrt{3}e}{2}\right)$

C. $\left(-k, -\frac{\sqrt{3}e}{2}\right)$

D. $\left(k_1 - \frac{\sqrt{3}e}{2}\right)$

Answer: D

Solution:

The equation of the hyperbola is

$$x^2 - ky^2 = 3$$

which can be rewritten in standard form as:

$$\frac{x^2}{3} - \frac{y^2}{\frac{3}{k}} = 1$$

Here, $a^2 = 3$ and $b^2 = \frac{3}{k}$.

The angle between the asymptotes is given by

$$\tan^{-1} \frac{b}{a} = \frac{\pi}{3}$$

This implies:

$$\tan \frac{\pi}{6} = \frac{b}{a} = \frac{\sqrt{\frac{3}{k}}}{\sqrt{3}} = \frac{1}{\sqrt{k}}$$

Therefore:

$$\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{k}} \Rightarrow k = 3$$

Recalculate the standard form with $k = 3$:

$$3x^2 - y^2 = 9$$

The pole of the line $x + y - 1 = 0$ with respect to the hyperbola $3x^2 - y^2 = 9$ can be found. Let (h, k) be the pole. The equation of the pole is defined using $S_1 = 0$:

$$3hx - ky = 9$$

This equation is rewritten as:

$$\frac{hx}{3} - \frac{ky}{3} = 1$$

Comparing with the equation $x + y - 1 = 0$, we get:

$$\frac{h}{3} = 1 \quad \text{and} \quad \frac{-k}{3} = 1$$

Solving these gives:

$$h = 3 \quad \text{and} \quad k = -9$$

Thus, the pole of the line is $(3, -9)$. The final expression representing the pole in terms of a parameter is:

$$\left(k, \frac{-\sqrt{3}e}{2}\right)$$

where $k = 3$.

Question39

The locus of point of intersection of tangents at the ends of normal chord of the hyperbola $x^2 - y^2 = a^2$ is

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Options:

A. $y^4 - x^4 = 4a^2x^2y^2$

B. $y^2 - x^2 = 4a^2x^2y^2$

C. $a^2(y^2 - x^2) = 4x^2y^2$

D. $y^2 + x^2 = 4a^2x^2y^2$

Answer: C

Solution:

Let $p(h, k)$ be the point of intersection of tangents at the ends of a normal chord of the hyperbola, $x^2 - y^2 = a^2$, then the equation of the chord is

$$hx - ky = a^2 \quad \dots (i)$$

But it is normal chord. So, its equation must be of the form $x \cos \theta + y \cot \theta = 2a \quad \dots (ii)$

Eqs. (i) and (ii) represents the same line

$$\begin{aligned}\therefore \frac{\cos \theta}{h} &= \frac{\cot \theta}{-k} = \frac{2a}{a^2} \\ \sec \theta &= \frac{a}{2h} \cdot \tan \theta = \frac{-a}{2k} \\ \therefore \sec^2 \theta - \tan^2 \theta &= 1 \\ \Rightarrow \frac{a^2}{4h^2} - \frac{b^2}{4k^2} &= 1 \\ \Rightarrow a^2 (k^2 - h^2) &= 4h^2 k^2\end{aligned}$$

Therefore, the locus of $p(h, k)$ is $a^2 (y^2 - x^2) = 4x^2 y^2$

Question40

If e_1 and e_2 are the eccentricities of the hyperbola $16x^2 - 9y^2 = 1$ and its conjugate respectively. Then, $3e_1 =$

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Options:

A. $5e_2$

B. $4e_2$

C. $2e_2$

D. e_2

Answer: B

Solution:

Given, hyperbola is $16x^2 - 9y^2 = 1$

$$\begin{aligned}\frac{x^2}{\frac{1}{16}} - \frac{y^2}{\frac{1}{9}} &= 1 \\ \Rightarrow \frac{x^2}{\left(\frac{1}{4}\right)^2} - \frac{y^2}{\left(\frac{1}{3}\right)^2} &= 1\end{aligned}$$

We know that Eccentricity of a hyperbola, $e_1^2 = 1 + \frac{b^2}{a^2}$

$$\Rightarrow e_1^2 = 1 + \frac{1 \times 16}{9 \times 1}$$

$$\Rightarrow e_1^2 = \frac{25}{9}$$

$$\Rightarrow e_1 = \frac{5}{3} \quad \dots (i)$$

Now, eccentricity of its conjugate

$$e_2^2 = 1 + \frac{a^2}{b^2}$$

$$\Rightarrow e_2^2 = 1 + \frac{1 \times 9}{16 \times 1}$$

$$\Rightarrow e_2^2 = \frac{25}{16}$$

$$\Rightarrow e_2 = \frac{5}{4}$$

From Eqs. (i) and (ii),

$$3e_1 = 3 \times \frac{5}{3} = \frac{5}{4} \times 4$$

$$\Rightarrow 3e_1 = 4e_2 \quad \left[\because e_2 = \frac{5}{4} \right]$$

Question41

If the normal to the rectangular hyperbola $x^2 - y^2 = 1$ at the point $P(\pi/4)$ meets the curve again at $Q(\theta)$, then $\sec^2 \theta + \tan \theta =$

AP EAPCET 2022 - 4th July Evening Shift

Options:

- A. 43
- B. 57
- C. 3
- D. 1

Answer: B

Solution:

Let $P(\sec \theta, \tan \theta)$ be the point where the normal to the rectangular hyperbola $x^2 - y^2 = 1$ Meet equation of normal at $\frac{\pi}{4}$ is

$$\begin{aligned}\frac{x}{\sec \frac{\pi}{4}} + \frac{y}{\tan \frac{\pi}{4}} &= 2 \\ \Rightarrow \frac{x}{\sqrt{2}} + y &= 2 \\ \Rightarrow x + \sqrt{2}y &= 2\sqrt{2}\end{aligned}$$

Substitute $x = 2\sqrt{2} - \sqrt{2}y$ in hyperbola's equation,

$$\begin{aligned}[\sqrt{2}(2 - y)]^2 - y^2 &= 1 \\ \Rightarrow 2[4 + y^2 - 4y] - y^2 &= 1 \\ \Rightarrow 8 + 2y^2 - 8y - y^2 - 1 &= 0 \\ \Rightarrow y^2 - 8y + 7 &= 0 \\ \Rightarrow y &= 7, 1 \\ x + 7\sqrt{2} &= 2\sqrt{2} \text{ or } x + \sqrt{2} = 2\sqrt{2} \\ \Rightarrow x &= -5\sqrt{2} \text{ or } x = \sqrt{2}\end{aligned}$$

The two points are $(-5\sqrt{2}, 7), (\sqrt{2}, 1)$.

At $(\frac{\pi}{4})$, the point is $(\sqrt{2}, 1)$ and at θ , the point is $(-5\sqrt{2}, 7)$.

$$\begin{aligned}\sec \theta &= -5\sqrt{2}, \tan \theta = 7 \\ \sec^2 \theta + \tan \theta &= (-5\sqrt{2})^2 + 7 = 57\end{aligned}$$

Question42

If the vertices and foci of a hyperbola are respectively $(\pm 3, 0)$ and $(\pm 4, 0)$, then the parametric equations of that hyperbola are

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Options:

- A. $x = 3 \sec \theta, y = 7 \tan \theta$
- B. $x = \sqrt{3} \sec \theta, y = \sqrt{7} \tan \theta$
- C. $x = \sqrt{3} \sec \theta, y = 7 \tan \theta$
- D. $x = 3 \sec \theta, y = \sqrt{7} \tan \theta$

Answer: D

Solution:

Vertices and foci of the hyperbola are $(\pm 3, 0)$ and $(\pm 4, 0)$.

Thus, $a = \pm 3$ and $ae = \pm 4$ and $e = \frac{4}{3}$

We know, $e = \sqrt{1 + \frac{b^2}{a^2}}$

$$\frac{16}{9} = 1 + \frac{b^2}{9} \Rightarrow b^2 = 7$$

Thus, equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{7} = 1$

whose parametric equation is $x = 3 \sec \theta$ and $y = \sqrt{7} \tan \theta$

Question43

The value of $\frac{1+\tanh x}{1-\tanh x}$ is

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Options:

A. e^x

B. e^{-2x}

C. e^{2x}

D. e^{-x}

Answer: C

Solution:

$$\begin{aligned}
& \frac{1 + \tanh x}{1 - \tanh x} \\
&= \frac{1 + \frac{e^x - e^{-x}}{e^x + e^{-x}}}{1 - \frac{e^x - e^{-x}}{e^x + e^{-x}}} \left[\because \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \right] \\
&= \frac{e^x + e^{-x} + e^x - e^{-x}}{e^x + e^{-x} - e^x + e^{-x}} \\
&= \frac{2e^x}{2e^{-x}} = e^{2x}
\end{aligned}$$

Question44

Let origin be the centre, $(\pm 3, 0)$ be the foci and $\frac{3}{2}$ be the eccentricity of a hyperbola. Then, the line $2x - y - 1 = 0$

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Options:

- A. intersects the hyperbola at two points.
- B. does not intersect the hyperbola.
- C. touches the hyperbola.
- D. passes through the vertex of the hyperbola.

Answer: B

Solution:

Hyperbola: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Centre : $O(0, 0)$

Foci : $S(\pm ae, o) \equiv (\pm 3, 0)$

$\Rightarrow ae = 3$

Eccentricity, $e = \sqrt{1 + \left(\frac{b}{a}\right)^2} = \frac{3}{2}$

$$\therefore ae = 3$$

$$\Rightarrow a \cdot \left(\frac{3}{2}\right) = 3$$

$$\Rightarrow a = 2$$

$$\text{and } \sqrt{1 + \left(\frac{b}{a}\right)^2} = \frac{3}{2}$$

$$\Rightarrow 1 + \left(\frac{b}{2}\right)^2 = \frac{9}{4}$$

$$\Rightarrow \frac{b^2}{4} = \frac{5}{4}$$

$$\Rightarrow b^2 = 5$$

$$\therefore \text{Hyperbola equation is } \frac{x^2}{4} - \frac{y^2}{5} = 1 \quad \dots (i)$$

$$\text{Given, line is } 2x - y = 1$$

Put $y = 2x - 1$ into Eq. (i),

$$\frac{x^2}{4} - \frac{(2x - 1)^2}{5} = 1$$

$$\Rightarrow 5x^2 - 4(4x^2 + 1 - 4x) = 20$$

$$\Rightarrow 5x^2 - 16x^2 - 4 + 16x - 20 = 0$$

$$\Rightarrow -11x^2 + 16x - 24 = 0$$

$$\text{Here, } D = b^2 - 4ac = (16)^2 - 4(-11)(-24) < 0$$

$\therefore$ Line does not intersect the hyperbola.

Question 45

The locus of a variable point whose chord of contact w.r.t. the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ subtends a right angle at the origin is

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Options:

A. $\frac{x^2}{4a^2} - \frac{y^2}{4b^2} = 1$

B. $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = \frac{x^2}{a^4} + \frac{y^2}{b^4}$

$$C. \frac{x}{a} - \frac{y}{b} = \frac{1}{a^2} + \frac{1}{b^2}$$

$$D. \frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2} - \frac{1}{b^2}$$

Answer: D

Solution:

Equation of hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \dots (i)$$

Let (h, k) be the point whose chord of contact subtends a right angle at the centre of hyperbola. Equation of chord of contact

$$\frac{hx}{a^2} - \frac{ky}{b^2} = 1$$

Squaring both the sides, we get

$$\begin{aligned} & \left(\frac{hx}{a^2} \right)^2 + \left(\frac{ky}{b^2} \right)^2 - 2 \frac{hx}{a^2} \cdot \frac{ky}{b^2} = (1)^2 \\ \Rightarrow & \frac{h^2 x^2}{a^4} + \frac{k^2 y^2}{b^4} - \frac{2hk}{a^2 b^2} \cdot xy = 1 \quad \dots (ii) \end{aligned}$$

Now, from Eqs. (i) and (ii), we have

$$\begin{aligned} & \frac{h^2 x^2}{a^4} + \frac{k^2 y^2}{b^4} - \frac{2hk}{a^2 b^2} \cdot xy = \frac{x^2}{a^2} - \frac{y^2}{b^2} \\ \Rightarrow & \left(\frac{h^2}{a^4} - \frac{1}{a^2} \right) x^2 + \left(\frac{k^2}{b^4} + \frac{1}{b^2} \right) y^2 - \frac{2hk}{a^2 b^2} \cdot (xy) = 0 \end{aligned}$$

The above equation represents a pair of perpendicular lines if coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow \frac{h^2}{a^4} - \frac{1}{a^2} + \frac{k^2}{b^4} + \frac{1}{b^2} = 0$$

$$\Rightarrow \frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{a^2} - \frac{1}{b^2}$$

Replacing $h \rightarrow x$ and $k \rightarrow y$,

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2} - \frac{1}{b^2}$$

Question46

If the focal chord of the hyperbola subtends a right angle at the center, then its eccentricity is

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Options:

A. $e = \frac{\sqrt{3}-1}{2}$

B. $e = \frac{\sqrt{5}-1}{2}$

C. $e = \frac{\sqrt{5}+1}{2}$

D. $e = \frac{\sqrt{3}+1}{2}$

Answer: C

Solution:

Let focal chord be latusrectum then it is given that it subtends 90° at its centre.

Image

Here, equation of hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \dots\dots (i)$$

So eccentricity, $e = \sqrt{1 + \frac{b^2}{a^2}}$

End points of latus rectum, $L = \left(ae, \frac{b^2}{a} \right)$ and $L' = \left(ae, -\frac{b^2}{a} \right)$

Centre $(0, 0)$

$$\therefore \angle LCL' = 90^\circ$$

In $\triangle LCL'$ by Pythagoras theorem

$$(LC)^2 + (L'C)^2 = (LL')^2$$

$$(ae - 0)^2 + \left(\frac{b^2}{a^2} - 0 \right)^2 + (ae - 0)^2 + \left(-\frac{b^2}{a} - 0 \right)^2 = \left(\frac{2b^2}{a} \right)^2$$

$$\Rightarrow 2 \left(a^2 e^2 + \frac{b^4}{a^2} \right) = \frac{4b^4}{a^2} \Rightarrow b^4 = a^4 e^2$$

$$\Rightarrow (e^2 - 1)^2 = e^2 \quad [\because b^2 = a^2 (e^2 - 1)]$$

$$\Rightarrow e^2 \pm e - 1 = 0$$

$$\Rightarrow e = \pm \frac{1 \pm \sqrt{1+4}}{2} = \frac{\pm 1 \pm \sqrt{5}}{2}$$

$\therefore e > 1$ for hyperbola

$$\Rightarrow e = \frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2} \Rightarrow e = \frac{\sqrt{5}+1}{2}$$

$\Rightarrow$ Option (c) is correct.

Question47

If one focus of a hyperbola is $(3, 0)$, the equation of its directrix is $4x - 3y - 3 = 0$ and its eccentricity $e = 5/4$, then the coordinates of its vertex is

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Options:

A. $\left(\frac{3}{5}, \frac{11}{5}\right)$

B. $\left(\frac{11}{5}, \frac{3}{5}\right)$

C. $\left(\frac{7}{5}, \frac{4}{5}\right)$

D. $\left(\frac{4}{5}, \frac{7}{5}\right)$

Answer: B

Solution:

Distance from focus to directrix is $a \left(e - \frac{1}{e}\right)$

$$a \left(\frac{5}{4} - \frac{4}{5}\right) = \frac{4 \cdot 3 - 3}{|5|} \Rightarrow a = 4$$

Slope of axis = $(-3/4)$

Distance from vertex to focus

$$= ae - a = a \left(\frac{5}{4}\right) - a = \frac{a}{4}$$

$$\begin{aligned} \text{Vertex} &= \left[3 \pm \left(\frac{-4}{5}\right), 0 \pm \left(\frac{3}{5}\right)\right] \\ &= \left(3 - \frac{4}{5}, \frac{3}{5}\right) = \left(\frac{11}{5}, \frac{3}{5}\right) \end{aligned}$$

Question48

The asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with any tangent to the hyperbola form a triangle whose area is $a^2 \tan(\alpha)$. Then, its eccentricity equals

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Options:

- A. $\sec(\alpha)$
- B. $\operatorname{cosec}(\alpha)$
- C. $\sec^2(\alpha)$
- D. $\operatorname{cosec}^2(\alpha)$

Answer: A

Solution:

Any tangent to hyperbola forms a triangle with asymptotes which has constant area $= ab$

$$\Rightarrow ab = a^2 \tan \alpha \Rightarrow \frac{b}{a} = \tan \alpha$$

$$\text{Eccentricity, } e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \tan^2 \alpha} = \sec \alpha$$
